

Dialysis

Definition

Dialysis is the diffusion of solutes from an area of **higher concentration** → **lower concentration** through a **semipermeable membrane**.

It is used to remove waste products from the blood in **renal failure**, especially **uraemia**.

Uraemia

Occurs when **>75% of nephrons are damaged** and is characterized by:

- ↑ **Nitrogenous waste products** in blood
- **Metabolic acidosis**
- **Hyperkalaemia**

Types of Dialysis

1. **Haemodialysis (artificial kidney)**
2. **Peritoneal dialysis**

1. Haemodialysis / Artificial Kidney

- Usually performed in hospital for **3–5 hours** through an IV access.
- Blood is passed through a **long, coiled semipermeable cellophane tube** immersed in dialysing fluid.
- **Heparin** is used as an anticoagulant.
- About **500 mL blood** is present in the machine at a time.
- Dialysed blood is returned to the patient through a **peripheral vein**.

Dialysing fluid

Its composition is similar to plasma but:

- **Free of waste products** → urea, uric acid, etc.
- ↓ **Na⁺, K⁺ and Cl⁻** compared with uraemic blood

- ↑ **Glucose, HCO_3^- and Ca^{2+}**

Mechanism

The cellophane membrane:

- Allows diffusion of **small dissolved substances**
- Does **not allow plasma proteins** to pass

Therefore:

Urea + other toxic wastes → blood → dialysing fluid

At the same time, abnormal electrolyte levels are corrected.

- Severe uraemia → usually **3 times/week**
- Can save life in **acute renal failure**
- Can prolong life in **chronic renal failure**

Important: Dialysis only **partially replaces the excretory function** of kidneys; it does **not replace** their **endocrine and metabolic functions**.

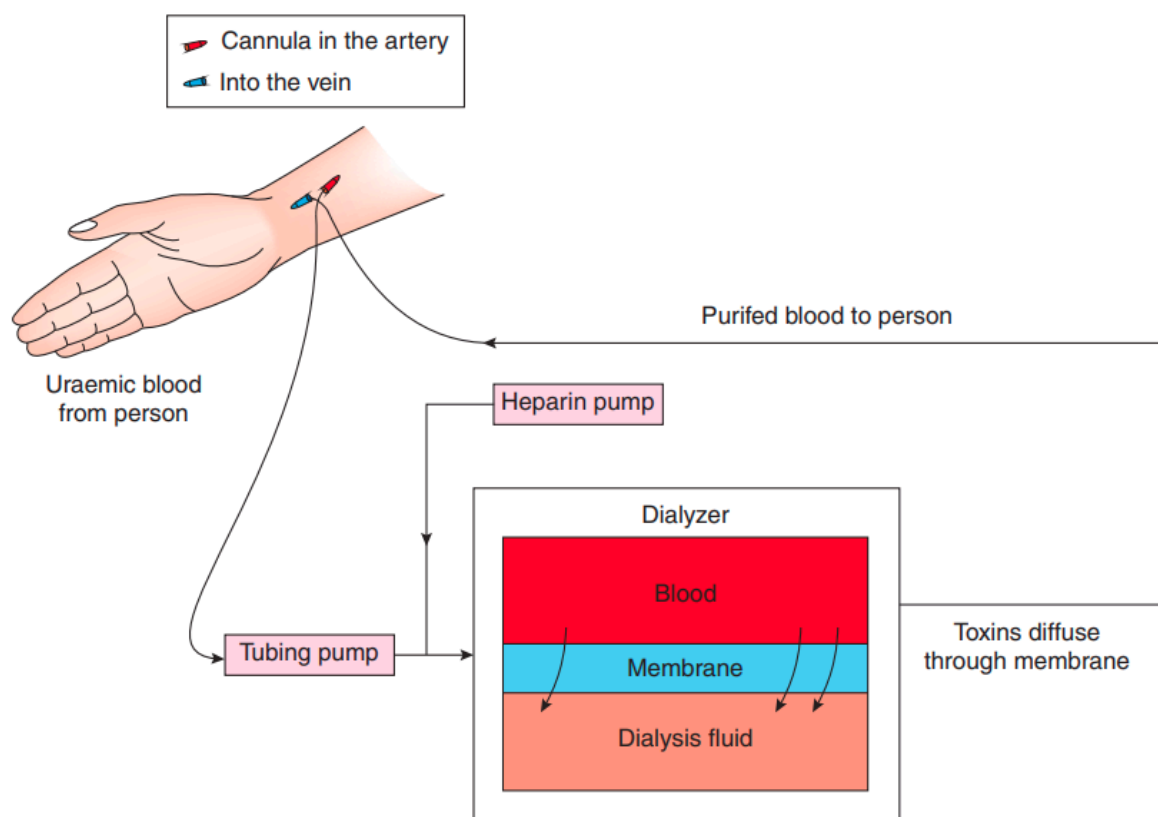


Fig. 6.6-4 Basic principle of haemodialysis.

2. Peritoneal Dialysis

- **Peritoneum acts as the semipermeable membrane.**
- About **2 L dialysing fluid** is introduced into the peritoneal cavity through an intraperitoneal catheter.
- Fluid remains for **15–20 min (dwell time)** → exchange occurs.
- Fluid is then drained and measured.
- One such process = **one cycle**.
- Usually performed at **6-hour intervals** → **4 cycles/day**.
- Can be performed **at home/work**, so hospitalization is not required.
- Useful in **young children and elderly patients with cardiovascular disorders**.
- Prolongs survival in chronic renal failure.

Renal Transplantation

Definition

Renal transplantation is the surgical transplantation of a healthy kidney into a patient with **chronic renal failure**. It is the **definitive treatment** for chronic renal failure.

Important Points

1. Donor

- Graft may be obtained from:
 - **Cadaver donor**
 - **Sibling**
 - **Parent**
 - **Twin** (*best donor*)

2. Site of transplantation

- Usually, the **left kidney of the donor** is transplanted into the **right iliac fossa** of the recipient.

3. Effect

- Reverses the **metabolic and excretory abnormalities** of chronic renal failure.
- Provides functional renal replacement and allows **complete rehabilitation**.

4. Immunosuppression

- Long-term immunosuppressive therapy is required to prevent rejection.
- Examples: **Prednisolone and cyclosporin**.

5. Advantage

- Offers **complete rehabilitation** and is considered a **cost-effective** treatment for chronic renal failure.

🔥 If asked as a short question:

Renal transplantation = definitive treatment of chronic renal failure → kidney from cadaver/relative → usually left donor kidney to right iliac fossa → lifelong immunosuppression → reverses metabolic & excretory abnormalities.

Last-minute recall

Dialysis → 2 types

Haemodialysis → artificial kidney → cellophane membrane

Peritoneal dialysis → peritoneum acts as membrane

Uraemia → >75% nephron damage → ↑ nitrogenous wastes + acidosis + hyperkalaemia

Dialysis does NOT replace → endocrine + metabolic functions of kidney